



Microsoft's speech recognition

Microsoft announces that its speech recognition system has achieved a new milestone putting it on par with humans. <http://dgit.in/SpeechAI>



A chip that repairs tissue

A new kind of skin-zapping chip can reprogram skin cells to act like other types of cells and repair damaged tissues. <http://dgit.in/SkinZap>

IARC team intends to inject the microbes into Venus' atmosphere. Two approaches are being considered here. A spray-trap mechanism where the microbes will simply be sprayed while the nano-spacecraft is in orbit. And the other mechanism being considered is emptying the tank of microbes all at once. The mechanism hasn't been finalised yet. The success of the Bijayan Venus Mission will be determined after the microbes have been injected into the atmosphere.

The payload needs to ensure the survivability of the microbes while in transit to Venus which is also the secondary mission challenge. The quantity of microbes to be sent hasn't been finalised and it relies on the type of the spacecraft. It's dependent on the total mission cost where the payload capacity will be adjusted accordingly. Pushkar stated that even 1 kg of microbes will be enough as they will multiply in favourable conditions.



Tardigrades can survive in the most extreme living conditions

THE NEXT STEP

With the mission objective fulfilled, the microbes may or may not survive in the harsh environment. If they do, a future mission will be planned to retrieve those microbes. The selected microbes are going to be genetically marked before they are sent to Venus. This will make it easy to identify whether they were the same microbes that we had sent. The chances are quite low for Venus to harbour microbial life as it's already been classified as a Category II object according to COSPAR standards. But still, as a precaution, they will be marked to help us differentiate them from any possible Venusian life forms. The Bijayan microbes will be detectable using spectral analysis if they manage to flourish.

THE BIGGEST HURDLES TO CROSS

Apart from building the different components of the mission including the launch vehicle, payload and instruments, Bijayan faces a few challenges. The possibility of initiating the terraforming comes after the extremophiles surviving the extremely hot atmosphere in the first place. Even if the high radiation levels and high pressure can be endured by our tough micro test subjects, the insanely high surface temperature might just sabotage the mission. But back in 2014, the Venus Express probe returned some fascinating data about the poles, where at a distance of 130 to 140 kilometres from the surface, the temperatures were much colder than Earth. The lowest recorded temperature was about minus 157 degrees Celsius, a piece of cake for the tough tardigrades as they are capable of surviving at temperatures as low as minus 272 degrees Celsius. But again, they can't survive in such extreme conditions for a long period. Since this temperature zone is located near the surface, we don't know whether the microbes will survive the extreme heat as they pass through the atmosphere. Then again, targeting the polar regions with a different dispersion mechanism, ensuring that it survives the freefall might just work out.

The idea of terraforming other planets has been under scrutiny for decades and it has been surrounded by a lot of debate on ethics. Dealing with the ethics of terraforming is a completely different ballpark that argues whether we should be playing God by affecting the natural processes of other planets. As a precaution for research purposes, it completely makes sense not to disturb planetary systems where microbial or any form of life exists. Several planets out there might support a different form of life unknown to us, and allowing another life form to peacefully co-exist is something many critics have encouraged.

When spacecrafts or astronauts are sent to space, they need to be sterilised before blasting off. We don't want to send off Earthly-microorganisms to other worldly bodies if we are seeking answers to chemical evolution or origins

of life on that body. These microbes might just explosively proliferate in an alien environment, which will be detrimental to our own research. This is phenomenon is called forward contamination. And this could be a possibility on Venus. Contamination is a serious issue when it comes to space exploration. In order to avoid any grave consequences, the Bijayan team has made it clear that they'll be only sending off non-pathogenic microbes. The Committee on Space Research (COSPAR) has a planetary protection policy across five categories to prevent biological contamination in outer space exploration missions. NASA also has a dedicated position, called the Planetary Protection Officer, responsible of ensuring there's none (ideally) or minimal contamination on their spacecrafts going for outer space missions. The United Nations Outer Space Treaty covers contamination under Article IX along with other frameworks to govern a responsible and peaceful exploration of outer space. It's like an international space law that has been ratified or officially signed by India.

The Bijayan team will soon be releasing a white paper of their project considering the UNOOSA (United Nations Office for Outer Space Affairs) treaties and COSPAR guidelines by the end of this year. Since Bijayan is an outer space mission, there is lesser framework required compared to Earth-orbit satellite missions. Also, Bijayan is going to be a passenger on the rocket which means they don't have to deal with too much of the framework. The details of the framework including approvals by the Department of Space (India) will be addressed in the paper. This year ISRO received funding for a second Mars mission and a maiden voyage to Venus. ISRO also started accepting proposals for space-based experiments to study Venus, an orbiter mission that will probably happen after 2020. However, IARC hasn't sent any proposals to ISRO for the mission yet.

Slated for a 2023 launch, IARC is expected to release more information about the mission soon. 